

Lecture Note: 4 — Constants and Variables

Course Code: 25B11CI112 • Department of Computer Science & Engineering • JUIT

1. Classification of C Constants

In C, constants are classified into two broad categories:

- Primary Constants: Integer Constants, Real / Floating-point Constants, Character Constants.
- Secondary Constants: Array, Pointer, Structure (struct), Union, Enum, etc.

2. Numeric Constants Details

- Decimal integer: Digits 0–9. Default size: 2 or 4 bytes. Range: -32768 to 32767 (16-bit) or -2^{31} to $2^{31}-1$ (32-bit).
- Octal integer: Leading 0 followed by digits 0–7 (e.g. 076, 0127).
- Hexadecimal integer: Leading 0x or 0X followed by digits 0–9 and A–F (e.g. 0x24, 0x87A).
- Real / Float: Must have a decimal point or exponent form (e.g. +325.34, 3.6e+5).

Character Constants, Strings & Variables

1. Character & String Constants

- Character Constant: Single character enclosed in single quotes (e.g. 'A', 'c', '\$', '9'). Associated with ASCII values (A = 65, a = 97, 0 = 48).
- String Constant: Sequence of characters enclosed within double quotes (e.g. "JUIT Solan", "Dennis"). Automatically null-terminated with \0 by compiler.
- Symbolic Constant: Defined with preprocessor directive: #define MAX 100 or #define PI 3.14159.

2. Variable Declarations & Initialization

```
int studentCount = 65;  
float cgpa = 8.74;  
char grade = 'A';  
char campus[] = "JUIT Wagnaghat";
```